



Research article

Separable synchronous auxiliary model adaptive momentum estimation strategy for a time-varying system with colored noise from on-line measurements

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ABSTRACT

The primary focus of this article is to explore parameter estimation for time-varying systems affected by colored noise. Based on the attributes of the time-varying system with colored noise under investigation, the original system is separated and two different subsystems are reconstructed. To address the influence of the hidden variables in the system and the time-varying noise signal, we introduce auxiliary models into the reconstructed systems to achieve the separation and synchronization estimation of the time-varying parameters within the system. With the aim of achieving high-precision performance in estimating the time-varying parameters, a separable synchronous auxiliary model adaptive momentum algorithm is presented by introducing bias correction to the momentum and gradient square term and online parameter estimation is implemented. The proposed algorithm is used for estimating the time-varying output error moving average model to verify the performance. The results of simulation experiments illustrate the efficacy of the proposed method for estimating the time-varying system with colored noise. Additionally, the proposed algorithm is extended to a kind of direct current (DC) motor modeling parameter identification problem and shows good tracking performance.

1. Introduction

In the majority of control methodologies, it is assumed that an accurate model of the system under control is available, which prompts the exploration of parameter estimation and system identification [1–3]. Substantial investigations have been conducted in the fields of adaptive parameter estimation and adaptive control [4–6]. For instance, Jha et al. introduced a stochastic gradient algorithm and applied it to model the dynamic system of a boiler [7]. It is recognized that the initial advancements in adaptive parameter estimation relied on the utilization of either recursive least squares (RLS) methods or gradient descent algorithms [8]. Many traditional identification algorithms have been well used to solve the problem of parameter identification with constant parameters [9–12]. For example, Ding et al. introduced the multi-innovation identification theory into the parameter identification by extending the single innovation vector to multiple innovation matrix and realized the interactive operation of data sampling and iterative estimation [13]. Zhang et al. introduced an adaptive filtering algorithm grounded in fractional-order principles, which innovation tackled the challenge encountered by conventional adaptive filters when dealing with intricate interference [14]. However, the adaptive estimation methods based on traditional algorithms are primarily applicable when

parameters of system to be estimated remain constant [15]. The performance of these estimation methods might not remain satisfactory when the parameters to be identified exhibiting temporal variability.

The examination of time-varying characteristics is imperative in addressing numerous challenges across engineering domains [16]. Research on system identification methods for time-varying systems has made significant progress in recent years. The introduction of the forgetting factor in the traditional algorithm and the application of multi-innovation identification methods improve the estimation accuracy of time-varying system parameters in simple noise environment [17]. To explore further characteristic of time-varying systems under complex noise environment, the parameter estimation of time-varying systems with colored noise has garnered increasing attention within the control community recently. Ji et al. proposed a gradient iterative algorithm for estimating time-varying system with colored noise, this algorithm considered the fluctuating patterns of system parameters and incorporated momentum to enable it to adaptively track the evolving parameters of the system [18]. Zhou et al. introduced a two-stage extended approach to locally linear radial basis function-based autoregressive models under conditions of colored noise [19]. However, the existing algorithms still have some limitations in the identification of time-varying systems

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with colored noise interference. For example, the estimation accuracy will degrade when the time-varying system is corrupted by complex noises. Meanwhile, improving real-time performance without sacrificing accuracy is an important research direction. Effectively combining deep learning with traditional identification methods to leverage their respective advantages is a research area worth further exploration.

Separable synchronous identification method was elucidated as a means of optimizing some signal parameter estimation problems with a special structure [20,21]. The separable gradient method address both nonlinear and linear optimization of parameter estimation problems where the objective functions involve a considerable number of parameters, which can be decomposed into several different manners [22]. Generally, depending on the characteristics of the parameters or models, the separable identification method can be categorized as either parameter separable and model separable [23]. This method has the potential to streamline algorithm complexity and often used to simplify large and complex models [24]. The auxiliary model identification method is an effective tool to estimate the unknown variables in the information vectors. The basic idea of the auxiliary model identification is to replace the unknowns variables by the output of the auxiliary model [25]. The auxiliary model is typically employed to address the issue of latent variables within the identification problem [26]. Based on the characteristics of the system to be identified, an auxiliary model is devised to substitute the pertinent variables in the original system [27,28]. This approach effectively resolves the challenge posed by unmeasured variables in the original system. Building upon these methodologies, we extend the concepts of separable identification and auxiliary model identification to the realm of identifying time-varying systems under colored noise conditions, this leads to the proposal of a novel method for estimating time-varying parameter systems with colored noise.

Furthermore, the adaptive moment (ADAM) estimation strategy amalgamates the strengths of the momentum and the adaptive learning rate for efficient parameter optimization in neural networks. The inception of the ADAM estimation strategy arose from the need to address challenges encountered by traditional gradient algorithms when dealing with varying parameter gradients. By amalgamating the advantages of the momentum and the root mean square propagation (RMS) method, the ADAM algorithm dynamically adjusts the learning rate for each parameter, enhancing estimation accuracy and overall performance [29]. The ADAM method demonstrates robustness against noise interference by dynamically adjusting the first-order momentum and second-order momentum throughout the iteration process, incorporating the average gradients [30]. Inspired by this observation, this article presents a novel adaptive estimation technique specifically designed for time-varying systems affected by colored noise. Differing from the conventional gradient or RLS algorithms mentioned previously, the separable synchronous auxiliary model adaptive momentum (SS-AM-ADAM) algorithm is proposed by combining the ADAM optimization strategy with the ideas of the separable identification and the auxiliary model identification. The proposed algorithm can simultaneously identify the noise characteristics under the premise of high accuracy estimation of the time-varying parameters. Comparative simulations between the proposed methods and traditional gradient techniques demonstrate that this new estimation approach achieves superior performance. The primary contributions of this paper are outlined as follows.

- Through the combination of the separable synchronous identification and the auxiliary model identification, an identification technology for time-varying systems with colored noise is proposed. The latent variables in the systems can be estimated synchronously.
- By introducing the adaptive momentum optimization strategy, the separable synchronous auxiliary model adaptive momentum identification algorithm is proposed, which realizes the combination of the traditional gradient algorithm and the deep learning optimization strategy.

- The proposed algorithm performs multiple iterations on the data of a single sampling point and make full use of the data information of the sampling points. The estimation accuracy of the time-varying parameters under the colored noise is greatly improved.

The structure of this article unfolds as follows. In Section 2, the issue concerning the identification model of time-varying systems is delineated. In Section 3, the separable synchronous auxiliary model method is given and a traditional extended gradient algorithm is derived. Section 4 deduces a SS-AM-ADAM algorithm. Section 5 provides the simulation results of the proposed algorithm. Ultimately, the conclusions are outlined in Section 6.

2. Problem description

In practical applications, many systems are susceptible to external factors, environmental changes, or inherent system dynamics, resulting in time-varying characteristics in parameters of systems and temporal variations in the statistical properties of noise. The output error type (OET) model with the ability to better capture the dynamic behavior of the system are known to be excellent candidates for this task. Therefore, the following time-varying output error moving average (OEMA) model parameter prediction strategies will be explored in this article.

The time-varying OEMA system with colored noise is expressed as

$$y_o(\hat{h}) = \frac{\sum_{j=1}^{n_\gamma} \gamma_j^r(\hat{h})}{1 + \sum_{j=1}^{n_\alpha} \alpha_j^r(\hat{h})} u^i(\hat{h} - j) + \rho(\hat{h}), \quad (1)$$

where $y_o(\hat{h})$ is the output of the system, $u^i(\hat{h})$ is the input of the system, $y_o(\hat{h}) = 0$, $u^i(\hat{h}) = 0$ as $\hat{h} < 0$. $\alpha_j^r(\hat{h})$ ($j = 1, 2, \dots, n_\alpha$) and $\gamma_j^r(\hat{h})$ ($j = 1, 2, \dots, n_\gamma$) are the time-varying parameters of the system, $\rho(\hat{h}) := R(\hat{h}, p)v(\hat{h})$ donates the colored noise of the system, and $v(\hat{h})$ represents the approximation error, which can be interpreted as a random noise having a mean of zero. The $R(\hat{h}, p)$ represents a polynomial of the unit backward shift operator p^{-1} with the following mathematical descriptions:

$$R(\hat{h}, p) := 1 + \sum_{j=1}^{n_q} q_j^r(\hat{h})p^{-j}.$$

Thus, the colored noise $\rho(\hat{h})$ can be written as

$$\rho(\hat{h}) = \sum_{j=1}^{n_q} q_j^r(\hat{h})v(\hat{h} - j) + v(\hat{h}).$$

where the $u^i(\hat{h})$ is an intrinsic component of the system, inaccessible to direct measurement. Concurrently, the output $y_o(\hat{h})$ is the noisy measurement of $\varpi^r(\hat{h})$. The noise-free output is defined as

$$\varpi^r(\hat{h}) := \frac{\sum_{j=1}^{n_\gamma} \gamma_j^r(\hat{h})}{1 + \sum_{j=1}^{n_\alpha} \alpha_j^r(\hat{h})} u^i(\hat{h} - j) \in \mathbb{R}. \quad (2)$$

Supposing that the orders of parameters n_α, n_γ are known, let $n_1 := n_\alpha + n_\gamma$. Define the information vector $Y(\hat{h})$ and the parameter vector $\Xi(\hat{h})$ expressed as

$$\begin{aligned} Y(\hat{h}) &:= [-\varpi^r(\hat{h} - 1), \dots, -\varpi^r(\hat{h} - n_\alpha), u^i(\hat{h} - 1), \dots, \\ &\quad u^i(\hat{h} - n_\gamma), v(\hat{h} - 1), \dots, v(\hat{h} - n_q)]^T \in \mathbb{R}^{n_1 + n_q}, \\ \Xi(\hat{h}) &:= [\alpha_1^r(\hat{h}), \dots, \alpha_{n_\alpha}^r(\hat{h}), \gamma_1^r(\hat{h}), \dots, \gamma_{n_\gamma}^r(\hat{h}), q_1^r(\hat{h}), \dots, q_{n_q}^r(\hat{h})]^T \in \mathbb{R}^{n_1 + n_q}. \end{aligned}$$

Therefore, the expression for the identification model of the time-varying OEMA system with colored noise is as follows

$$y_o(\hat{h}) = Y^T(\hat{h})\Xi(\hat{h}) + v(\hat{h}). \quad (3)$$

The proposed parameter estimation algorithms in this paper are based on the parameter identification model in (3). Many identification methods are derived based on the identification models of systems [31–35] and can be used to estimate the parameters of other linear stochastic

systems, bilinear stochastic systems, nonlinear stochastic systems [36–40] and multivariable stochastic systems [41–45] and can be applied to other fields such as signal processing and process control systems. The application of the OET model in system identification contributes to a more accurate description of the dynamic characteristics and provides a clear framework for model parameter estimation and validation.

3. Separable synchronous auxiliary model identification

By observing the OEMA model, we can find that the model consists of two components: the noise-free output $\varpi^r(h)$ and the colored noise $\rho(h)$. The information vectors associated with these two components are intertwined, inevitably escalating the complexity of identification computations. We solve this problem by separating the information vector $Y(h)$ and the parameter vector $\Xi(h)$ based on the distinctive characteristics of these two information components.

Define the parameter vectors $\alpha(h)$ and $\zeta(h)$ and the information vectors $\rho(h)$ and $\psi(h)$ as

$$\rho(h) := [-\varpi^r(h-1), \dots, -\varpi^r(h-n_\alpha), u^i(h-1), \dots, u^i(h-n_\beta)]^T \in \mathbb{R}^{n_1},$$

$$\alpha(h) := [\alpha_1^r(h), \dots, \alpha_{n_\alpha}^r(h), \gamma_1^r(h), \dots, \gamma_{n_\gamma}^r(h)]^T \in \mathbb{R}^{n_1},$$

$$\psi(h) := [v(h-1), \dots, v(h-n_q)]^T \in \mathbb{R}^{n_q},$$

$$\zeta(h) := [q_1^r(h), \dots, q_{n_q}^r(h)]^T \in \mathbb{R}^{n_q}.$$

Meanwhile, the original time-varying OEMA system with colored noise is further decomposed into two subsystems as

$$y_{o1}(h) = \rho^T(h)\alpha(h) + v(h), \quad (4)$$

$$\rho(h) = \psi^T(h)\zeta(h) + v(h). \quad (5)$$

Subsequently, the identification model of the time-varying OEMA system can be formulated as

$$y_o(h) = \rho^T(h)\alpha(h) + \psi^T(h)\zeta(h) + v(h). \quad (6)$$

The two equivalent expressions in (4) and (5) can be written as

$$y_{o1}(h) = y_o(h) - \psi^T(h)\zeta(h), \quad (7)$$

$$\rho(h) = y_o(h) - \rho^T(h)\alpha(h). \quad (8)$$

Let $\hat{\alpha}(h) := [\hat{\alpha}_1^r(h), \dots, \hat{\alpha}_{n_\alpha}^r(h), \hat{\gamma}_1^r(h), \dots, \hat{\gamma}_{n_\gamma}^r(h)]^T \in \mathbb{R}^{n_1}$ and $\hat{\zeta}(h) := [\hat{q}_1^r(h), \dots, \hat{q}_{n_q}^r(h)]^T \in \mathbb{R}^{n_q}$ be the estimates of the information vectors $\alpha(h)$ and $\zeta(h)$.

Utilize the two sub-identification models introduced in (4) and (5) and formulate their corresponding criterion functions as

$$J_1(\hat{\alpha}(h)) := \frac{1}{h} \sum_{j=1}^h [y_{o1}(j) - \rho^T(j)\hat{\alpha}(j)]^2, \quad (9)$$

$$J_2(\hat{\zeta}(h)) := \frac{1}{h} \sum_{j=1}^h [\rho(j) - \psi^T(j)\hat{\zeta}(j)]^2. \quad (10)$$

Minimize criterion functions $J_1(\hat{\alpha}(h))$ and $J_2(\hat{\zeta}(h))$, the stochastic gradient (SG) algorithms of two subsystems can be formulated as

$$\hat{\alpha}(h) = \hat{\alpha}(h-1) + \tau_1 \hat{\rho}(h)[y_{o1}(h) - \rho^T(h)\hat{\alpha}(h-1)], \quad (11)$$

$$\hat{\zeta}(h) = \hat{\zeta}(h-1) + \tau_2 \hat{\psi}(h)[\rho(h) - \psi^T(h)\hat{\zeta}(h-1)], \quad (12)$$

where τ_j ($j = 1, 2$) denote learning rates, which are pivotal hyperparameters within gradient descent algorithms, serving to determine the magnitude of parameter updates at each iteration. The gradient descent is a fundamental optimization algorithm and endeavors to minimize a specified loss function by iteratively adjusting the model parameters in the direction of the gradient of the objective function [46–49].

According to Eq. (6), we can find that two subsystems are associated. Based on Eqs. (4) and (5) and the gradient search, we have the extended stochastic gradient (ESG) algorithm expressed as

$$\hat{\alpha}(h) = \hat{\alpha}(h-1) + \tau_1 \rho(h)[y_o(h) - \rho^T(h)\hat{\alpha}(h-1) - \psi^T(h)\hat{\zeta}(h-1)], \quad (13)$$

$$\hat{\zeta}(h) = \hat{\zeta}(h-1) + \tau_2 \psi(h)[y_o(h) - \rho^T(h)\hat{\alpha}(h-1) - \psi^T(h)\hat{\zeta}(h-1)]. \quad (14)$$

According to the structure of Model (2), since the system input $u^i(h)$ is difficult to be directly measured as a hidden variable, we introduce an auxiliary model to substitute for the output without noise. According to the auxiliary model identification idea [50–53], letting $\hat{\varpi}_a^r$ be the output of the auxiliary model and using $\hat{\varpi}_a^r(h-j)$ replace $\varpi^r(h-j)$ in information vector $\rho(h)$, we get the estimated information vector

$$\hat{\rho}(h) := [-\hat{\varpi}_a^r(h-1), -\hat{\varpi}_a^r(h-2), \dots, -\hat{\varpi}_a^r(h-n_\alpha), u^i(h-1), u^i(h-2), \dots, u^i(h-n_\gamma)]^T \in \mathbb{R}^{n_1}.$$

The auxiliary model $\hat{\varpi}_a^r(t)$ can be formulated as

$$\hat{\varpi}_a^r(h) := \hat{\rho}^T(h)\hat{\delta}(h) \quad (15)$$

In practice, the colored noise present in the system is not directly measurable, which pose a challenge to the identification process. In this article, the noise terms $v(h-j)$ are unknown. Define the auxiliary model for computing $\hat{v}_a(h)$ as

$$\hat{v}_a(h) := y_o(h) - \hat{\rho}^T(h)\alpha(h) - \psi^T(h)\zeta(h). \quad (16)$$

Define the estimate of information vector $\psi(h)$ expressed as

$$\hat{\psi}(h) := [\hat{v}_a(h-1), \hat{v}_a(h-2), \dots, \hat{v}_a(h-n_q)]^T \in \mathbb{R}^{n_q}.$$

Based on the replaced information vectors, we redefine the negative gradient expressed as

$$g_1 := -\hat{\rho}(h)[y_o(h) - \hat{\rho}^T(h)\hat{\delta}(h-1) - \hat{\psi}^T(h)\hat{\zeta}(h-1)], \quad (17)$$

$$g_2 := -\hat{\psi}(h)[y_o(h) - \hat{\rho}^T(h)\hat{\delta}(h-1) - \hat{\psi}^T(h)\hat{\zeta}(h-1)], \quad (18)$$

Taking negative gradients into the ESG algorithm, the auxiliary model extended stochastic gradient (AM-ESG) algorithm can be expressed as

$$\hat{\delta}(h) = \hat{\delta}(h-1) - \tau_1 g_1, \quad (19)$$

$$\hat{\zeta}(h) = \hat{\zeta}(h-1) - \tau_2 g_2. \quad (20)$$

The parameter vectors $\alpha(h)$ and $\zeta(h)$ contain all the parameters of the system to be identified. Nevertheless, only input–output data $u^i(h)$ and $y_o(h)$ are available, while the information vectors $\rho(h)$ and $\psi(h)$ within $Y(h)$ remain unknown. because intermediate variable $\varpi^r(h)$ unknown. Hence, it is imperative to construct auxiliary models for estimating these elusive variables using the concept of the auxiliary model identification. The proposed algorithm in this paper can combine other identification algorithms [54–59] to explore new parameter identification methods for other linear systems and nonlinear systems [60–65] and can be applied to engineering systems [66–70] such as information process and process control systems. Algorithm 1 lists the steps of computing the parameter estimation vectors $\hat{\delta}(h)$ and $\hat{\zeta}(h)$ using the AM-ESG algorithm.

4. The SS-AM-ADAM algorithm

Traditional gradient descent algorithms determine the magnitude of model parameter updates solely based on the current gradient. These methods rely exclusively on first derivative information to determine the next search direction. Moreover, the convergence rate of traditional gradient descent algorithms tends to be slow. To address this issue and ensure the accuracy of parameter estimation, we denote by h_s the s iterations at time h , redefine the gradient as

$$g_{1,s} := -\hat{\rho}(h_s)[y_o(h_s) - \hat{\rho}^T(h_s)\hat{\delta}(h_s-1) - \hat{\psi}^T(h_s)\hat{\zeta}(h_s-1)], \quad (21)$$

$$g_{2,s} := -\hat{\psi}(h_s)[y_o(h_s) - \hat{\rho}^T(h_s)\hat{\delta}(h_s-1) - \hat{\psi}^T(h_s)\hat{\zeta}(h_s-1)], \quad (22)$$

where s denotes the number of iterations for the current dataset and incorporate momentum into the adjustment of the parameter matrix during iteration.

Suppose that at the current iteration step s , then the update rule for momentum is as follows:

$$m_{1,s} = \lambda_1 m_{1,s-1} + (1 - \lambda_1) g_{1,s}, \quad (23)$$

$$m_{2,s} = \lambda_2 m_{2,s-1} + (1 - \lambda_2) g_{2,s}. \quad (24)$$

Algorithm 1 The AM-ESG algorithm

```

1: Initialize
   Initialize:  $\hat{h} = 0$ ;
   Let  $p_0 = 10^6$ ;
   Set the learn rate  $\tau_j$ ;
   Set the data length  $M$ ;
   Set the starting value  $\hat{o}(\hat{h}) = \mathbf{1}_{n_1}/p_0$ ;
   Set the starting value  $\hat{\zeta}(\hat{h}) = \mathbf{1}_{n_g}/p_0$ ;
2: Estimate the parameter vectors  $\hat{o}(\hat{h})$  and  $\hat{\zeta}(\hat{h})$ .
3: for  $\hat{h} = 0, 1, 2, \dots, M$  do
4:   Collect  $u^i(\hat{h})$  and  $y_o(\hat{h})$ .
5:   Compute the  $g_1$  by (17).
6:   Compute the  $g_2$  by (18).
7:   Update the estimation vector  $\hat{o}(\hat{h})$  using (19).
8:   Update the estimation vector  $\hat{\zeta}(\hat{h})$  using (20).
9:   Construct the auxiliary model  $\hat{\phi}_a^r(\hat{h})$  by (15).
10:  Construct the auxiliary model  $\hat{v}_a(\hat{h})$  by (16).
11:  if  $\hat{h} < M$ 
12:     $\hat{h} = \hat{h} + 1$ ;
13:  else
14:    Obtain  $\hat{\Xi}(\hat{h})$ , break
15:  end if
16: end for

```

Where λ_j ($j = 1, 2$) donate the momentum factors, which govern the weighting assigned to historical gradients during the computation of momentum, thereby dictating the direction and magnitude of parameter updates.

Remark 1. The momentum factor typically takes values in the range of 0 to 1, often approaching 1. Rigorous experimentation and empirical adjustments are typically necessary to ascertain the most suitable momentum factor for a given optimization problem.

The introduction of the momentum term in the current iteration update serves to alleviate gradient corrections at the current moment, enabling the algorithm to employ a larger step size for quicker convergence while reducing oscillations.

The RMSP algorithm is an optimization method designed to address the challenge of learning rate selection in deep learning. The RMSP maintains an exponentially weighted moving average of the squared gradients to estimate the variations in gradients for each parameter, the update rule of the RMSP is expressed as:

$$\kappa_{1,s} = \beta_1 \kappa_{1,s-1} + (1 - \beta_1) g_1 \odot g_1, \quad (25)$$

$$\kappa_{2,s} = \beta_2 \kappa_{2,s-1} + (1 - \beta_2) g_2 \odot g_2, \quad (26)$$

where β_j ($j = 1, 2$) are squared gradient factors and $0 \leq \beta_j < 1$. $\odot$ means that the elements in the matrix correspond to the multiplication.

Remark 2. The RMSP has adaptive learning rate mechanism, which shows good performance when faced with non-stationary data or gradients of varying magnitudes.

During the initial stages of the algorithm, moment estimates may suffer from substantial biases, potentially leading to overly optimistic or pessimistic learning rate calculations. An optimistic learning rate might result in overly rapid parameter updates, while a pessimistic one might lead to slow learning. Introducing bias correction helps the algorithm more accurately adapt to changes in gradients and promotes stable and faster convergence towards suitable parameter values.

Define the bias correction items of the momentum:

$$\bar{m}_1 = \frac{m_{1,s}}{(1 - \lambda_1^s)}, \quad (27)$$

$$\bar{m}_2 = \frac{m_{2,s}}{(1 - \lambda_2^s)}. \quad (28)$$

By implementing this corrective measure, the estimate of the momentum term becomes more accurate. This enhancement significantly contributes to improved algorithmic stability.

Remark 3. The paramount significance of the bias correction for the momentum term lies in the mitigation of biases, particularly during the nascent phases of the algorithm and when initializing the momentum term to 0. This correction is instrumental in ensuring that the estimate of momentum term is not unduly influenced by its initialization, thereby providing a more precise reflection of gradient information.

Similarly to the momentum term, during the initial iterations, the update rule for the bias-corrected squared gradient term $\bar{\kappa}_j$ ($j = 1, 2$) are expressed as:

$$\bar{\kappa}_1 = \frac{\kappa_{1,s}}{(1 - \gamma_1^s)}, \quad (29)$$

$$\bar{\kappa}_2 = \frac{\kappa_{2,s}}{(1 - \gamma_2^s)}. \quad (30)$$

The bias correction for the gradient squared term is crucial for fortifying algorithm stability. This correction promotes a more accurate reflection of gradient information during the initial training phases, facilitating accelerated convergence towards global optima in optimization algorithms.

According to (23)–(30). Utilize the bias-corrected momentum and squared gradient terms to update the model parameter $\hat{o}(\hat{t}_s)$ and $\hat{\zeta}(\hat{t}_s)$, we have get the SS-AM-ADAM algorithm expressed as

$$\hat{o}(\hat{h}_s) = \mu_1 \bar{m}_1 \oslash (\sqrt{\bar{\kappa}_1} + \epsilon), \quad (31)$$

$$\hat{\zeta}(\hat{h}_s) = \mu_2 \bar{m}_2 \oslash (\sqrt{\bar{\kappa}_2} + \epsilon). \quad (32)$$

where ϵ is a small constant vector added for numerical stability. In theoretical terms, this value approaches 0. Here, $\oslash$ represents the division of the preceding matrix element by the subsequent matrix element.

Algorithm 2 lists the steps of computing the parameter estimation vectors $\hat{o}(\hat{h}_s)$ and $\hat{\zeta}(\hat{h}_s)$ using the SS-AM-ADAM algorithm. The presented parameter estimation algorithm in this paper can joint other estimation strategies [71–76] be applied to many engineering science areas to model dynamic systems and information processing systems [77–85] for obtaining the characteristic parameters of process control systems [86–95].

5. Examples

5.1. Example 1

Consider the following OEMA time-varying system with colored noise:

$$y_o(t_s) = \frac{\sum_{j=1}^1 \alpha_j^r(\hat{h}_s)}{\sum_{j=1}^1 \gamma_j^r(\hat{h}_s)} u^i(\hat{h}_s - j) + \phi(\hat{h}_s),$$

$$\phi(t_s) = q_1^r(\hat{h}_s) v(\hat{h}_s - 1).$$

the time-varying parameters $\alpha_1^r(\hat{h}_s)$, $\gamma_1^r(\hat{h}_s)$ and $q_1^r(\hat{h}_s)$ are measured as

$$\alpha_1^r(\hat{h}_s) = 0.0003\hat{h}_s - 0.021\sqrt{2\hat{h}_s + 5},$$

$$\gamma_1^r(\hat{h}_s) = -0.13 \cos(0.001\pi\hat{h}_s),$$

$$q_1^r(\hat{h}_s) = 0.15 - 0.016\sqrt{0.1\hat{h}_s + 0.002}.$$

and their changing trends are shown in Fig. 1.

During simulation, the input $u^i(\hat{h}_s)$ is generated as an uncorrelated uniform distribution random signal sequence with a zero mean and unit variance. Meanwhile, $v(\hat{h}_s)$ is generated as a normally distributed white noise sequence with a zero mean and variance σ^2 . Setting the data length $M = 1300$ and learn rate $\tau_j = 0.01$ ($j = 1, 2$) the input-output data $u^i(\hat{h}_s)$, $y_o(\hat{h}_s)$ are shown in Fig. 2 and the time-varying parameters are shown in Fig. 1. Applying the AM-ESG algorithm and the input-output data $u^i(\hat{h}_s)$, $y_o(\hat{h}_s)$ to estimate the parameters of this example

Table 1The AM-ESG and SS-AM-ADAM estimates and their errors with variance $\sigma^2 = 0.50^2$.

Algorithms	h_s	$\alpha'_1(h_s)$	$\hat{\alpha}'_1(h_s)$	$\gamma'_1(h_s)$	$\hat{\gamma}'_1(h_s)$	$q'_1(h_s)$	$\hat{q}'_1(h_s)$	$\delta(\%)$
AM-ESG	100	-0.2741	-0.1164	-0.1233	-0.1174	0.0986	-0.0110	60.7502
	300	-0.4298	-0.2855	-0.0744	-0.0749	0.0615	-0.0200	37.6356
	500	-0.5179	-0.3197	0.0025	0.0028	0.0362	-0.0239	39.8905
	800	-0.6020	-0.5027	0.1059	0.1540	0.0066	-0.0366	19.3917
	1000	-0.6413	-0.6116	0.1300	0.1306	-0.0105	-0.0403	6.4382
	1300	-0.6830	-0.6546	0.0727	0.0769	-0.0332	-0.0444	4.4823
SS-AM-ADAM	100	-0.2741	-0.2648	-0.1233	-0.1226	0.0986	0.0991	2.9592
	300	-0.4298	-0.4226	-0.0744	-0.0734	0.0615	0.0529	2.5546
	500	-0.5179	-0.5438	0.0025	0.0090	0.0362	0.0542	6.2070
	800	-0.6020	-0.6027	0.1059	0.1054	0.0066	0.0074	0.1923
	1000	0.6413	-0.6394	0.1300	0.1299	-0.0105	-0.0167	0.9735
	1300	-0.6830	-0.6843	0.0727	0.0728	-0.0332	-0.0413	1.1880

Table 2The AM-ESG and SS-AM-ADAM estimates and their errors with variance $\sigma^2 = 0.30^2$.

Algorithms	h_s	$\alpha'_1(h_s)$	$\hat{\alpha}'_1(h_s)$	$\gamma'_1(h_s)$	$\hat{\gamma}'_1(h_s)$	$q'_1(h_s)$	$\hat{q}'_1(h_s)$	$\delta(\%)$
AM-ESG	100	-0.2741	-0.1164	-0.1233	-0.1174	0.0986	-0.0110	60.7516
	300	-0.4298	-0.2855	-0.0744	-0.0749	0.0615	-0.0200	37.6311
	500	-0.5179	-0.3197	0.0025	0.0024	0.0362	-0.0239	39.8848
	800	-0.6020	-0.5026	0.1059	0.1542	0.0066	-0.0366	19.4072
	1000	-0.6413	-0.6116	0.1300	0.1306	-0.0105	-0.0403	6.4367
	1300	-0.6830	-0.6546	0.0727	0.0770	-0.0332	-0.0444	4.4866
SS-AM-ADAM	100	-0.2741	-0.2663	-0.1233	-0.1238	0.0987	0.1021	2.7236
	300	-0.4298	-0.4256	-0.0744	-0.0745	0.0615	0.0957	7.8264
	500	-0.5179	-0.5077	0.0025	0.0023	0.0362	0.0742	7.5846
	800	-0.6020	-0.6001	0.1059	0.1057	0.0066	0.0348	4.6225
	1000	-0.6413	-0.6407	0.1300	0.1299	-0.0105	0.0108	3.2537
	1300	-0.6830	-0.6852	0.0727	0.0728	-0.0332	-0.0153	2.6182

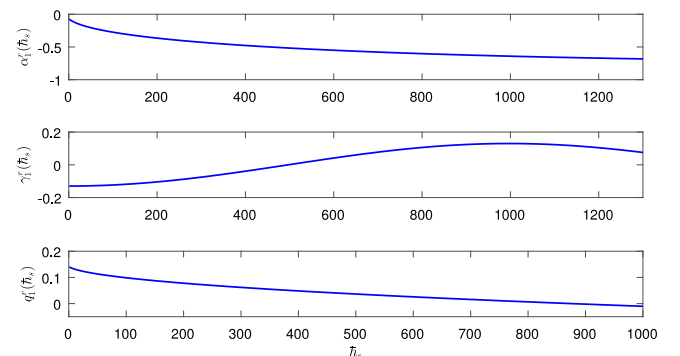
Table 3The AM-ESG and SS-AM-ADAM estimates and their errors with variance $\sigma^2 = 0.10^2$.

Algorithms	h_s	$\alpha'_1(h_s)$	$\hat{\alpha}'_1(h_s)$	$\gamma'_1(h_s)$	$\hat{\gamma}'_1(h_s)$	$q'_1(h_s)$	$\hat{q}'_1(h_s)$	$\delta(\%)$
AM-ESG	100	-0.2741	-0.11640	-0.1233	-0.1173	0.0986	-0.0110	60.7513
	300	-0.4298	-0.2855	-0.0744	-0.0750	0.06150	-0.0200	37.6287
	500	-0.5179	-0.3198	0.0025	0.0020	0.0362	-0.0239	39.8776
	800	-0.6020	-0.5025	0.1059	0.1543	0.0066	-0.0365	19.4248
	1000	-0.6413	-0.6116	0.1300	0.1306	-0.0105	-0.0403	6.4345
	1300	-0.6830	-0.6546	0.0727	0.0770	-0.0332	-0.0444	4.4901
SS-AM-ADAM	100	-0.27414	-0.26780	-0.12325	-0.12369	0.09864	0.10391	2.61024
	300	-0.42983	-0.42457	-0.07441	-0.07375	0.06148	0.05358	2.16007
	500	-0.51789	-0.51327	0.00245	0.00357	0.03618	0.03991	1.16513
	800	-0.60198	-0.60231	0.10589	0.10571	0.00662	0.00360	0.49775
	1000	-0.64133	-0.63980	0.12998	0.13034	-0.01048	-0.01004	0.24917
	1300	-0.6830	-0.68359	0.07273	0.07270	-0.03320	-0.03371	0.11061

system. Utilizing the simulation model parameters along with the input signal, the output signal $y_o(h_s)$ is generated. The AM-ESG parameter estimates and errors versus $\delta := \|\hat{o}(h_s) - o(h_s)\| / \|o(h_s)\| \times 100\%$ with different noise variances are shown in Tables 1 to 3. Utilizing the SS-AM-ADAM algorithm along with the input-output data $u^i(h_s), y_o(h_s)$ for estimating the parameters of this OEMA system, the SS-AM-ADAM parameter estimates and errors across different noise variances are depicted in Tables 1 to 3 and Fig. 3.

In order to showcase the advantages of the proposed SS-AM-ADAM algorithm, under noise variance $\sigma^2 = 0.10^2$, we use the AM-ESG algorithm with the learn rate $\tau_j = 0.01$ ($j = 1, 2$) and the SS-AM-ADAM algorithm with the hyperparameters $\lambda_j = 0.90$, $\beta_j = 0.99$ ($j = 1, 2$) to identify this example OEMA time-varying system, the parameter estimation errors δ for two algorithms are shown in Fig. 4. Fig. 5 shows the condition of parameter estimation. Figs. 6 to 8 illustrate the system output, model output, and their respective errors over time for the two algorithms. Parameter estimation errors for eleven randomly selected data points within the data set length are displayed in Fig. 9.

From Tables 1 to 3, based on the analysis, the following conclusions can be drawn: as the noise levels decrease, the SS-AM-ADAM algorithm exhibits enhanced accuracy in parameter estimation compared to the AM-ESG algorithm. This is apparent from the parameter estimation

**Fig. 1.** The time-varying parameters.

errors and the parameter estimation error curves illustrated in Fig. 3. From Fig. 4, at equivalent noise levels, the SS-AM-ADAM algorithm provides more precise parameter estimates compared to the AM-ESG algorithm.

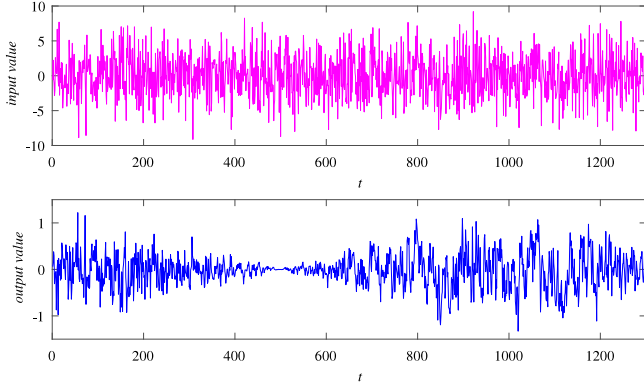
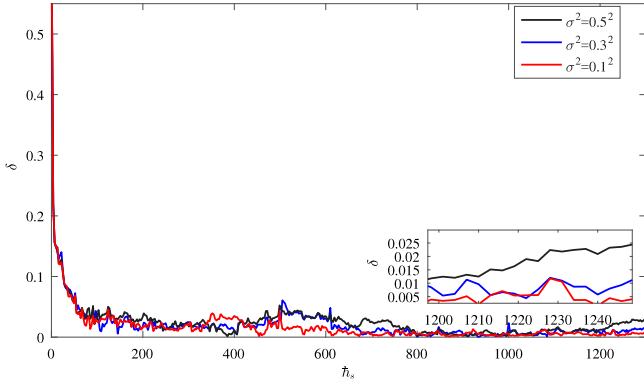
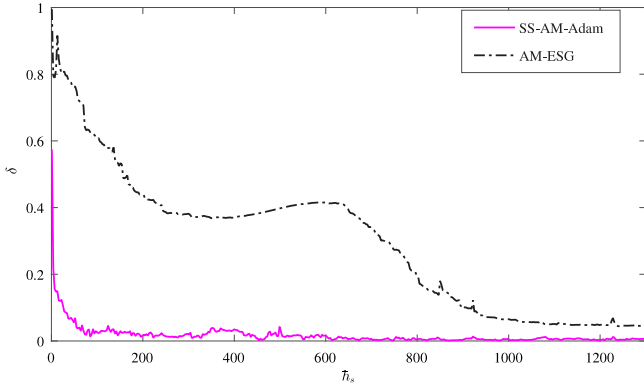
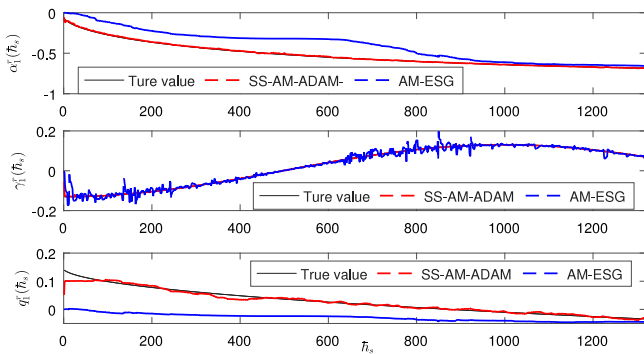
Fig. 2. The input-output data $u^i(h_s), y_o(h_s)$.

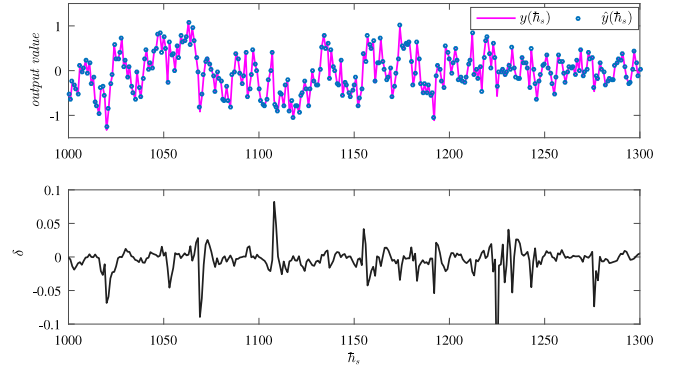
Fig. 3. The parameter estimates and their errors with different noise variances for the SS-AM-ADAM algorithm.

Fig. 4. The parameter estimation error δ versus h_s for two algorithms.Fig. 5. The parameter estimation versus h_s for two algorithms.**Algorithm 2** The SS-AM-ADAM algorithm

```

1: Initialize
   Initialize:  $\hat{h}_s = 0$ ;
   Set the momentum factors  $\lambda_j$ ;
   Set the squared gradient factors  $\beta_j$ ;
   Set the momentum  $\mathbf{m}_i$ ;
   Set the gradient squared term  $\kappa_i$ ;
   Set the bias in the momentum term  $\bar{\mathbf{m}}_i$ ;
   Set the bias-corrected squared gradient term  $\bar{\kappa}_j$ ;
   Set the data length  $M$ ;
   Let  $p_0 = 10^6$ ;  $\epsilon = 10^{-8}$ ;
   Set the starting value  $\hat{o}(h_s) = \mathbf{1}_{n_1}/p_0$ ;
   Set the starting value  $\hat{\zeta}(h_s) = \mathbf{1}_{n_d}/p_0$ ;
2: Estimate the parameter vectors  $\hat{o}$  and  $\hat{\zeta}$ .
3: for  $h_s = 0, 1, 2, \dots, M$  do
4:   Collect  $u^i(h_s)$  and  $y_o(h_s)$ .
5:   for  $s \leq s_{max}$  do
6:     Compute the  $\mathbf{g}_{1,s}$  by (17).
7:     Compute the  $\mathbf{g}_{2,s}$  by (18).
8:     Update the estimation vector  $\hat{o}(h_s)$  using (31).
9:     Update the estimation vector  $\hat{\zeta}(h_s)$  using (32).
10:    Construct the auxiliary model  $\hat{\omega}_a^r(h_s)$  by (15).
11:    Construct the auxiliary model  $\hat{\delta}_a(h_s)$  by (16).
12:   end for
13:   if  $h_s < M$ 
14:      $h_s = h_s + 1$ ;
15:   else
16:     Obtain  $\hat{o}$  and  $\hat{\zeta}$ , break
17:   end
18: end for

```

Fig. 6. The system output and the model output and their errors versus h_s for AM-ESG algorithm.**Table 4**

The average running time for different algorithms.

Algorithms	Average running time (s)
AM-ESG	0.01738
MIFG	0.02854
SS-AM-ADAM	0.02258

As shown in Fig. 5, the SS-AM-ADAM algorithm demonstrates the capability to track and estimate changes in time-varying parameters. In addition, we can also see that the SS-AM-ADAM algorithm have estimation accuracy to the true output value than AM-ESG algorithm — see the picture in Fig. 6 to Fig. 9.

In order to compare the estimation accuracy and efficiency of the proposed algorithm with other classical algorithms, apply the SG algorithm, the multi-innovation stochastic gradient algorithm (MISG) algorithm and the SS-AM-ADAM algorithm under noise variance $\sigma^2 =$

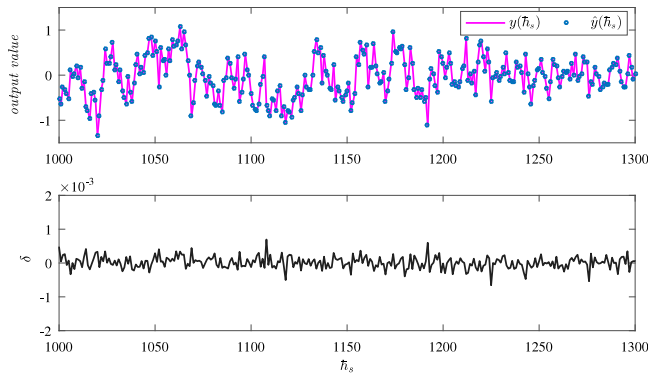


Fig. 7. The system output and the model output and their errors versus t for SS-AM-ADAM algorithm.

Table 5

The parameter estimates and their errors with different noise variances for the SS-AM-ADAM algorithm.

Symbols	Descriptions
$E(s)$	Armature potential (V)
$U(s)$	Motor terminal voltage (V)
$I(s)$	Armature current (A)
$T_m(s)$	Electromagnetic torque (N m)
K_r	Motor reverse electromotive force constant (Vs/rad)
$\Omega(s)$	Angular speed of motor (rad/s)
K_e	Motor torque constant (N m/A)
J	Rotor moment of inertia (kg m ²)
v	Armature impedance (Ω)
L	Coefficient of self inductance (H)
f	Hydraulic friction coefficient of the rotor

0.10^2 to identify the example OEMA time-varying system with colored noise. The parameter estimation errors δ for these algorithms are shown in Fig. 10. The Table 4 shows the average running time of the three algorithms when the estimation error is below 5%.

From Table 4 and Figs. 4 and 10, under the same conditions, although the SS-AM-ADAM algorithm runs faster than the MIFG algorithm but slightly slower than the AM-ESG algorithm, the accuracy of the SS-AM-ADAM algorithm is much higher than other algorithms, which is caused by the iterative process of the SS-AM-ADAM algorithm at each sampling point, which will increase the computational burden to a certain extent. However, with the development of computers, the accuracy benefits of the SS-AM-ADAM algorithm far outweigh the computational burden.

5.2. Example 2

The DC motor is roughly composed of three parts: permanent magnet motor, control circuit and position detection, in which the rotor is excited by permanent magnet and the stator has multi-phase winding. The mathematical model of the DC motor can be derived by combining the potential balance equation with the torque balance equation of the DC motor.

The transfer function of the DC motor can be established as.

$$\begin{cases} E(s) = K_e \Omega(s) = U(s) - vI(s) - LsI(s) \\ T_m(s) = K_t I(s) = Js\Omega(s) - f\Omega(s) \end{cases}$$

The physical meaning of the symbols in the discrete mathematical model of the DC motor is shown in Table 5.

Numerically, $K_e = K_t = \ell = 2NBlr$, where ℓ donates the coupling coefficient of the motor. Here N is the number of magnetic poles, l is

the length of the coil located in the magnetic field, B is the magnetic flux density, and r is the radius of the armature coil to the axis. After calculation, the transfer function of the system is derived as

$$H(s) = \frac{\Omega(p)}{U(p)} = \frac{k/LJ}{s^2 + \frac{rJ+Lf}{LJ}s + \frac{rf+\chi^2}{LJ}}.$$

After zero-order sampling and preserving discretization, it can be transformed into a transfer function expressed as

$$G(d) = (1 - d^{-1})Z \frac{H(s)}{s} = \frac{\gamma_1'(\hat{h}_s)d + \gamma_2'(\hat{h}_s)}{d^2 + \alpha_1'(\hat{h}_s)d + \alpha_2'(\hat{h}_s)}. \quad (33)$$

According to Eq. (33), the difference equation between the angular velocity $\zeta(\hat{h}_s)$ of the motor and the terminal voltage $\xi(\hat{h}_s)$ of the motor is established as

$$\zeta(\hat{h}_s) = \frac{\sum_{j=1}^2 \gamma_j'(\hat{h}_s)}{1 + \sum_{j=1}^2 \alpha_j'(\hat{h}_s)} \xi(\hat{h}_s - j) + \chi(\hat{h}_s).$$

In the given range of time-varying parameters, the parameter $\alpha_2'(\hat{h}_s) \approx 0$, can be ignored, so the model of the system is further simplified and expressed as

$$\zeta(\hat{h}_s) = \frac{\sum_{j=1}^2 \gamma_j'(\hat{h}_s)}{1 + \alpha_1'(\hat{h}_s)} \xi(\hat{h}_s - j) + \chi(\hat{h}_s). \quad (34)$$

Because the armature resistance has time-varying characteristics, the parameters in the system also have time-varying characteristics, the DC motor model time-varying parameters are show as Fig. 11.

Setting the data length $M = 1500$, applying the SS-AM-ADAM algorithm with the hyperparameters $\lambda_j = 0.90$, $\beta_j = 0.99$ ($j = 1, 2$) and the learn rate $\tau_j = 0.01$ ($j = 1, 2$) to identify this system. The first 400 groups of data of angular velocity $\zeta(\hat{h}_s)$ and terminal voltage $\xi(\hat{h}_s)$ are shown in Fig. 12, the parameter estimation for the SS-AM-ADAM algorithm are shown in Fig. 13. Applying the SS-AM-ADAM algorithm with different the noise variance σ , the SS-AM-ADAM parameter estimates and errors versus $\hat{h}_s$ with different noise variances are shown in Fig. 14.

From Figs. 13 and 14, we can draw the following conclusions. The proposed algorithm has a significant effect on DC motor parameter identification. As the noise levels decrease, the SS-AM-ADAM algorithm is capable of providing more precise parameter estimates and exhibits superior robustness.

6. Conclusions

This article investigates parameter estimation for a category of time-varying OEMA parametric systems subject to colored noise. Building upon the principles of the separable identification and the auxiliary model identification, the paper introduces the SS-AM-ADAM algorithm. Comparative analysis with traditional gradient algorithms demonstrates that the SS-AM-ADAM algorithm effectively enhances estimation accuracy and speed for time-varying systems under colored noise conditions. Simulation results confirm the efficacy of the proposed algorithm, which also yields promising outcomes in parameter identification for a DC motor model. Future research directions include exploring parameter estimation methods for other nonlinear or linear systems with colored noise, as well as fast time-varying systems, among others. The presented parameter estimation algorithm in this paper can combine some estimation algorithms for studying new methods for linear and nonlinear stochastic systems with colored noises and can be applied to control engineering fields, and transportation communication field [96–104] such as control and schedule systems [105–115] and the process control systems.

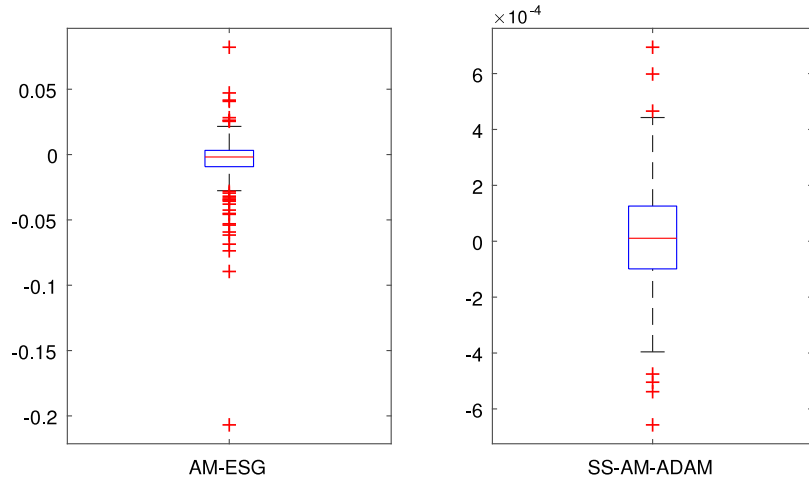


Fig. 8. The system output and the model output and their errors versus h_s for two algorithms.

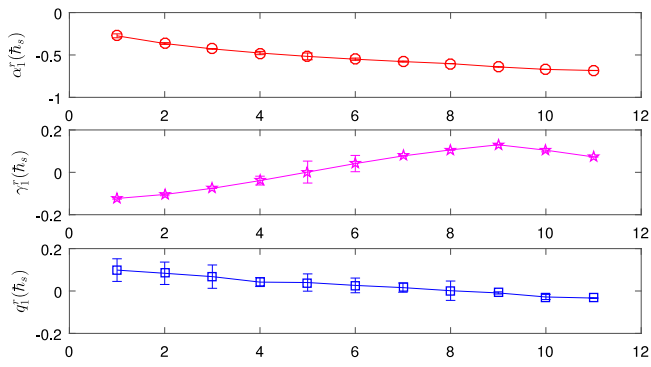


Fig. 9. The parameter estimation errors of the corresponding data points.

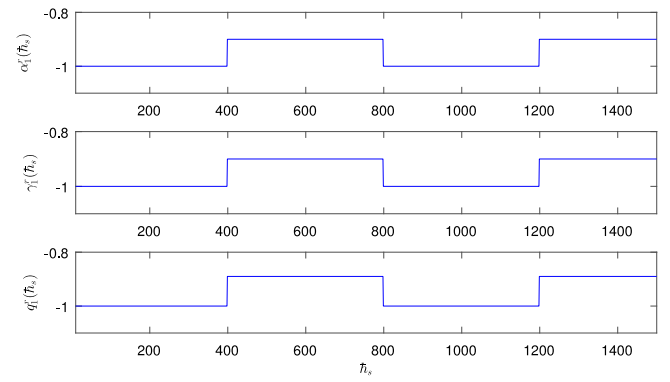


Fig. 11. The time-varying parameters.

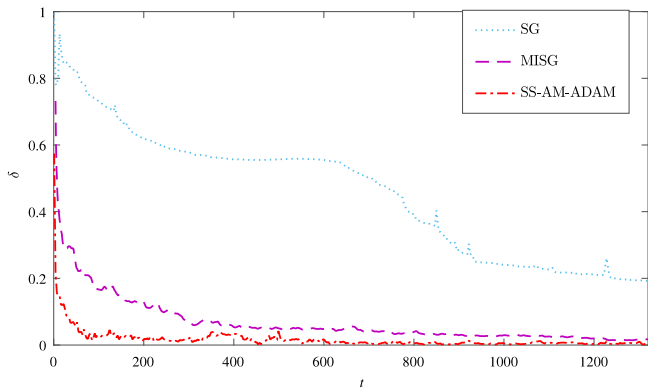


Fig. 10. The parameter estimation errors δ for different algorithms.

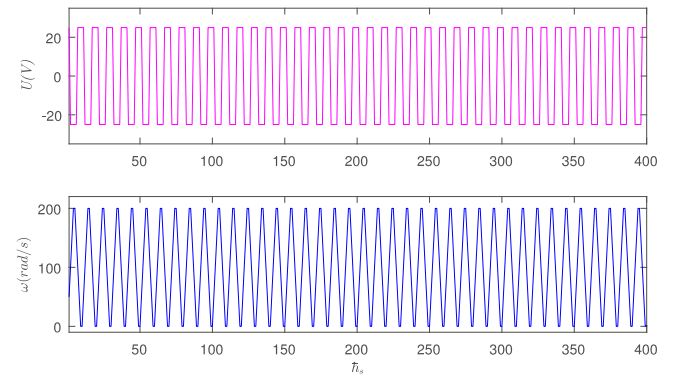


Fig. 12. The input-output data $\xi(h_s), \zeta(h_s)$.

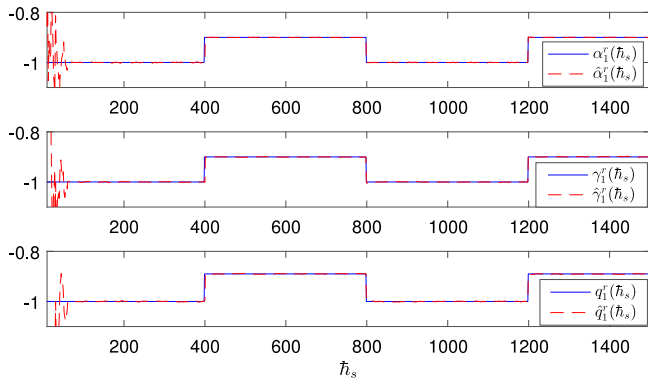


Fig. 13. The time-varying characteristics of time-varying parameters.

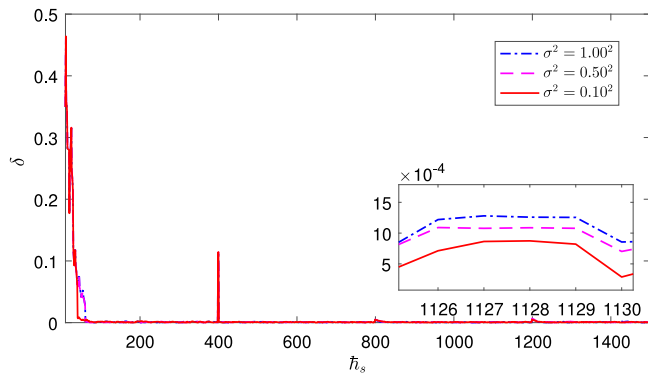


Fig. 14. The estimation errors δ versus h_s for different noise variances for the SS-AM-ADAM algorithm.

CRedit authorship contribution statement

Yanshuai Zhao: Writing – original draft. **Yan Ji:** Writing – review & editing, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Xu L. Separable Newton recursive estimation method through system responses based on dynamically discrete measurements with increasing data length. *Int J Control Autom Syst* 2022;20(2):432–43.
- [2] Wang YJ, Liu YT, Chen JH, Tang SH, Deng MQ. Online identification of Hammerstein systems with B-spline networks. *Internat J Adapt Control Signal Process* 2024;38(6):2074–92.
- [3] Xu L. Parameter estimation for nonlinear functions related to system responses. *Int J Control Autom Syst* 2023;21(6):1780–92.
- [4] Wang YJ, Yang L. An efficient recursive identification algorithm for multilinear systems based on tensor decomposition. *Internat J Robust Nonlinear Control* 2021;31(11):7920–36.
- [5] Liu X, Qin PL. Parameter estimation and model-free multi-innovation adaptive control algorithms. *Int J Control Autom Syst* 2024;22(11):3509–24.
- [6] Xu H, Xu L, Shen S. Online identification methods for a class of Hammerstein nonlinear systems using the adaptive particle filtering. *Chaos Solitons Fractals* 2024;86:115181.
- [7] Jha RS, Jha NN, Lele MM. Stochastic gradient descent algorithm for the predictive modelling of grate combustion and boiler dynamics. *ISA Trans* 2022;136:571–89.
- [8] Na J, Xing YS, Costa-Castelló R. Adaptive estimation of time-varying parameters with application to Roto-Magnet plant. *IEEE Trans Syst Man Cybern Syst* 2021;51(2):731–41.
- [9] Hou J. Parsimonious model based consistent subspace identification of Hammerstein systems under periodic disturbances. *Int J Control Autom Syst* 2024;22(1):61–71.
- [10] Hou J, Su H, Yu CP, Chen FW, Li PH. Bias-correction errors-in-variables Hammerstein model identification. *IEEE Trans Ind Electron* 2023;70(7):7268–79.
- [11] Gu Y, Hou YT, Li CJ, Zhu YF. Separable synchronous gradient-based iterative algorithms for the nonlinear ExpARX system. *Internat J Adapt Control Signal Process* 2024;38(12):3801–18.
- [12] Hou J, Su H, Yu CP, Chen FW, Li PH, Xie HF, Li TF. Consistent subspace identification of errors-in-variables Hammerstein systems. *IEEE Trans Syst Man Cybern: Syst* 2023;53(4):2292–303.
- [13] Ding F. Least squares parameter estimation and multi-innovation least squares methods for linear fitting problems from noisy data. *J Comput Appl Math* 2023;426:115107.
- [14] Zhang X. Optimal adaptive filtering algorithm by using the fractional-order derivative. *IEEE Signal Process Lett* 2022;29:399–403.
- [15] Liu WT, Xiong WL. Robust gradient estimation algorithm for a stochastic system with colored noise. *Int J Control Autom Syst* 2023;21(2):553–62.
- [16] Ji Y, Liu J, Liu HB. An identification algorithm of generalized time-varying systems based on the Taylor series expansion and applied to a pH process. *J Process Control* 2023;128:103007.
- [17] Zhu XG, Liu ST. On reachable set bounding for discrete-time switched nonlinear positive systems with mixed time-varying delays and disturbance. *Nonlinear Anal: Hybrid Syst* 2023;49:101373.
- [18] Ji Y, Jiang AN. Filtering-based accelerated estimation approach for generalized time-varying systems with disturbances and colored noises. *IEEE Trans Circuits Syst II Express Briefs* 2023;70(1):206–10.
- [19] Zhou YH, Yang EF. Two-stage extended recursive gradient algorithm for locally linear RBF-based autoregressive models with colored noises. *ISA Trans* 2022;129:284–94.
- [20] Ding F, Ma H, Pan J, Yang EF. Hierarchical gradient- and least squares-based iterative algorithms for input nonlinear output-error systems using the key term separation. *J Franklin Inst* 2021;358(9):5113–35.
- [21] Xu L. Separable synchronous multi-innovation gradient-based iterative signal modeling from on-line measurements. *IEEE Trans Instrum Meas* 2022;71:6501313.
- [22] Li JM. Synchronous optimization schemes for dynamic systems through the kernel-based nonlinear observer canonical form. *IEEE Trans Instrum Meas* 2022;71:3210952.
- [23] Ma H, Wang Y, Ji ZC. A novel three-stage quality oriented data-driven nonlinear industrial process monitoring strategy. *IEEE Trans Instrum Meas* 2022;71:3524711.
- [24] Gu Y, Dai W, Zhu QM, Nouri H. Hierarchical multi-innovation stochastic gradient identification algorithm for estimating a bilinear state-space model with moving average noise. *J Comput Appl Math* 2023;420:114794.
- [25] Ding F. *System identification-auxiliary model identification idea and methods*. Beijing: Science Press; 2017.
- [26] Liu J, Ji Y. Auxiliary model-based recursive least squares algorithm for two-input single-output Hammerstein output-error moving average systems by using the hierarchical identification principle. *Internat J Robust Nonlinear Control* 2022;32(8):7575–93.
- [27] Ding F, Chen T. Combined parameter and output estimation of dual-rate systems using an auxiliary model. *Automatica* 2004;40(10):1739–48.
- [28] Ding F, Chen T. Parameter estimation of dual-rate stochastic systems by using an output error method. *IEEE Trans Autom Control* 2005;50(9):1436–41.
- [29] Zhou YF, Zhang MC, Zhu JL, Zheng RJ, Wu QT. A Randomized Block-Coordinate Adam online learning optimization algorithm. *Neural Comput Appl* 2020;32(16):12671–84.
- [30] Shin Y, Darbon J, Karniadakis GE. Accelerating gradient descent and adam via fractional gradients. *Neural Netw* 2023;161:185–201.
- [31] Ding F, Xu L, Zhang X, Ma H. Hierarchical gradient- and least-squares-based iterative estimation algorithms for input-nonlinear output-error systems by using the over-parameterization. *Internat J Robust Nonlinear Control* 2024;34(2):1120–47.
- [32] Hu C, Liu HB, Ji Y. Parameter and order estimation algorithms and convergence analysis for lithium-ion batteries. *Internat J Robust Nonlinear Control* 2023;33(18):11411–33.
- [33] Hu C, Ji Y, Ma CQ. Joint two-stage multi-innovation recursive least squares parameter and fractional-order estimation algorithm for the fractional-order input nonlinear output-error autoregressive model. *Internat J Adapt Control Signal Process* 2023;37(7):1650–70.
- [34] Wang XY, Ma JX, Xiong WL. Expectation–maximization algorithm for bilinear state-space models with time-varying delays under non-Gaussian noise. *Internat J Adapt Control Signal Process* 2023;37(10):2706–24.
- [35] Sun SY, Xu L. Filtered multi-innovation-based iterative identification methods for multivariate equation-error ARMA systems. *Internat J Adapt Control Signal Process* 2023;37(3):836–55.
- [36] Wang YJ, Tang SH, Gu XB. Parameter estimation for nonlinear Volterra systems by using the multi-innovation identification theory and tensor decomposition. *J Franklin Inst* 2022;359(2):1782–802.

- [37] Sun SY, Wang X. Hierarchical iterative identification algorithms for a nonlinear system with dead-zone and saturation nonlinearity based on the auxiliary model. *Internat J Adapt Control Signal Process* 2023;37(7):1866–92.
- [38] Xu L. Auxiliary model multi-innovation stochastic gradient parameter estimation methods for nonlinear sandwich systems. *Internat J Robust Nonlinear Control* 2021;31(1):148–65.
- [39] Bi YQ, Ji Y. Parameter estimation of fractional-order Hammerstein state space system based on the extended Kalman filter. *Internat J Adapt Control Signal Process* 2023;37(7):1827–46.
- [40] Liu WX, Li MH. Unbiased recursive least squares identification methods for a class of nonlinear systems with irregularly missing data. *Internat J Adapt Control Signal Process* 2023;37(8):2247–75.
- [41] Ma H, Pan J, Ding W. Partially-coupled least squares based iterative parameter estimation for multi-variable output-error-like autoregressive moving average systems. *IET Control Theory Appl* 2019;13(18):3040–51.
- [42] Pan J, Ma H, Zhang X. Recursive coupled projection algorithms for multivariable output-error-like systems with coloured noises. *IET Signal Process* 2020;14(7):455–66.
- [43] Zhang X. Recursive parameter estimation methods and convergence analysis for a special class of nonlinear systems. *Internat J Robust Nonlinear Control* 2020;30(4):1373–93.
- [44] Xing HM, Yang EF. Hierarchical recursive least squares parameter estimation methods for multiple-input multiple-output systems by using the auxiliary models. *Internat J Adapt Control Signal Process* 2023;37(11):2983–3007.
- [45] Jin Y. A coupled recursive least squares algorithm for multivariable systems and its computational amount analysis by using the coupling identification concept. *Internat J Adapt Control Signal Process* 2024;38(2):513–33.
- [46] An S, He Y, Wang LJ. Maximum likelihood based multi-innovation stochastic gradient identification algorithms for bilinear stochastic systems with ARMA noise. *Internat J Adapt Control Signal Process* 2023;37(10):2690–705.
- [47] Xu L, Zhu QM. Novel parameter estimation method for the systems with colored noises by using the filtering identification idea. *Systems Control Lett* 2024;186:105774.
- [48] Xing HM, Yang EF. Highly-efficient filtered hierarchical identification algorithms for multiple-input multiple-output systems with colored noises. *Systems Control Lett* 2024;186:105762.
- [49] Xu L, Xu H. Adaptive multi-innovation gradient identification algorithms for a controlled autoregressive autoregressive moving average model. *Circuits Systems Signal Process* 2024;43(6):3718–47.
- [50] Ding F. Filtered auxiliary model recursive generalized extended parameter estimation methods for Box-Jenkins systems by means of the filtering identification idea. *Internat J Robust Nonlinear Control* 2023;33(10):5510–35.
- [51] Bedoui S, Abderrahim K. Iterative parameter identification for hammerstein systems with ARMA noises by using the filtering identification idea. *Internat J Adapt Control Signal Process* 2024;38(9):3134–60.
- [52] Ding F, Xu L, Zhang X, Zhou Y, Luan X. Recursive identification methods for general stochastic systems with colored noises by using the hierarchical identification principle and the filtering identification idea. *Annu Rev Control* 2024;57:100942.
- [53] Miao G, Yang D. Auxiliary model-based recursive least squares and stochastic gradient algorithms and convergence analysis for feedback nonlinear output-error systems. *Internat J Adapt Control Signal Process* 2024;38(10):3268–89.
- [54] Li JM, Hayat T. A novel nonlinear optimization method for fitting a noisy Gaussian activation function. *Internat J Adapt Control Signal Process* 2022;36(3):690–707.
- [55] Wang JW, Ji Y. Two-stage gradient-based iterative algorithms for the fractional-order nonlinear systems by using the hierarchical identification principle. *Internat J Adapt Control Signal Process* 2022;36(7):1778–96.
- [56] Fan YM, Liu XM. Auxiliary model-based multi-innovation recursive identification algorithms for an input nonlinear controlled autoregressive moving average system with variable-gain nonlinearity. *Internat J Adapt Control Signal Process* 2022;36(3):521–40.
- [57] Fan YM, Liu XM. Two-stage auxiliary model gradient-based iterative algorithm for the input nonlinear controlled autoregressive system with variable-gain nonlinearity. *Internat J Robust Nonlinear Control* 2020;30(14):5492–509.
- [58] Ding F, Shao X-L, Xu L, Zhang X, Xu H, Zhou YH. Filtered generalized iterative parameter identification for equation-error autoregressive models based on the filtering identification idea. *Internat J Adapt Control Signal Process* 2024;38(4):1363–85.
- [59] Yang D. Multi-innovation gradient-based iterative identification methods for feedback nonlinear systems by using the decomposition technique. *Internat J Robust Nonlinear Control* 2023;33(13):7755–73.
- [60] Yang D, Yang EF. Hierarchical gradient-based iterative parameter estimation algorithms for a nonlinear feedback system based on the hierarchical identification principle. *Circuits Systems Signal Process* 2024;43(1):124–51.
- [61] Miao GQ, Yang EF. Iterative parameter identification algorithms for transformed dynamic rational fraction input-output systems. *J Comput Appl Math* 2023;434:115297.
- [62] Xu L. Decomposition and composition modeling algorithms for control systems with colored noises. *Internat J Adapt Control Signal Process* 2024;38(1):255–78.
- [63] Ding J, Wu M, Xiao M. Nonlinear decoupling control with PI lambda Du neural network for MIMO systems. *IEEE Trans Neural Netw Learn Syst* 2022;35(6):8715–22.
- [64] Xu L. Separable synthesis estimation methods and convergence analysis for multivariable systems. *J Comput Appl Math* 2023;427:115104.
- [65] Liu QY, Chen FY. Model transformation based distributed stochastic gradient algorithm for multivariate output-error systems. *Int J Syst Sci* 2023;54(7):1484–502.
- [66] Zhou YH, Ling KV. Online network-based identification and its application in satellite attitude control systems. *IEEE Trans Aerosp Electron Syst* 2023;59(3):2530–43.
- [67] Wang YJ, Tang SH, Deng MQ. Modeling nonlinear systems using the tensor network B-spline and the multi-innovation identification theory. *Internat J Robust Nonlinear Control* 2022;32(13):7304–18.
- [68] Zhou YH. A novel coupled recursive multivariate nonlinear time-series modelling method by using interactive identification. *Appl Math Model* 2024;127:571–87.
- [69] Ding F, Chen T, Qiu L. Bias compensation based recursive least squares identification algorithm for MISO systems. *IEEE Trans Circuits Syst II Express Briefs* 2006;53(5):349–53.
- [70] Xu N. Convergence analysis of a synchronous gradient estimation scheme for time-varying parameter systems. *J Comput Appl Math* 2024;443:115724.
- [71] Li MH, Liu XM. Maximum likelihood hierarchical least squares-based iterative identification for dual-rate stochastic systems. *Internat J Adapt Control Signal Process* 2021;35(2):240–61.
- [72] Ding F, Yang HZ, Liu F. Performance analysis of stochastic gradient algorithms under weak conditions. *Sci China Ser F-Inf Sci* 2008;51(9):1269–80.
- [73] Li MH, Liu XM. Iterative identification methods for a class of bilinear systems by using the particle filtering technique. *Internat J Adapt Control Signal Process* 2021;35(10):2056–74.
- [74] Ding F, Liu XM, Ma XY. Kalman state filtering based least squares iterative parameter estimation for observer canonical state space systems using decomposition. *J Comput Appl Math* 2016;301:135–43.
- [75] Ibrahim K. Parameter estimation methods for time-invariant continuous-time systems from dynamical discrete output responses based on the Laplace transforms. *Internat J Adapt Control Signal Process* 2024;38(9):3213–32.
- [76] Xu H. Joint parameter and time-delay estimation for a class of nonlinear time-series models. *IEEE Signal Process Lett* 2022;29:947–51.
- [77] Luo K, Yang G, Li Y, Lan S, Wang Y, He L, Hu B. Croup and pertussis cough sound classification algorithm based on channel attention and multiscale Mel-spectrogram. *Biomed Signal Process Control* 2024;91:106073.
- [78] Yang G, He Z, Su Z, Li Y, Hu B. Keyframe recommendation based on feature intercross and fusion. *Complex Intell Syst* 2024;10(4):4955–71.
- [79] Xu J, Li Y, Yang G, He L, Luo K. Multiscale facial expression recognition based on dynamic global and static local attention. *IEEE Trans Affect Comput* 2024. <http://dx.doi.org/10.1109/TAFFC.2024.3458464>.
- [80] Xu CJ, Zeng W, Liu C, Yan HC. Event-triggered semi-global output consensus of discrete-time multi-agent systems with input saturation and external disturbances. *IEEE Trans Circuits Syst II: Express Briefs* 2023;70(12):4469–73.
- [81] Xu C, Qin Y, Su H. Observer-based dynamic event-triggered bipartite consensus of discrete-time multi-agent systems. *IEEE Trans Circuits Syst II Express Briefs* 2023;70(3):1054–8.
- [82] Li JM. Online parameter optimization scheme for the kernel function-based mixture models disturbed by colored noises. *IEEE Trans Circuits Syst II Express Briefs* 2024;71(8):3960–4.
- [83] Xu C, Xu H, Guan Z, Ge Y. Observer-based dynamic event-triggered semi-global bipartite consensus of linear multi-agent systems with input saturation. *IEEE Trans Cybern* 2023;53(5):3139–52.
- [84] Liu SY, Zhang X. Expectation-maximization algorithm for bilinear systems by using the Rauch-Tung-Striebel smoother. *Automatica* 2022;142:110365.
- [85] Liu SY, Wang YJ, Hayat T. Joint iterative state and parameter estimation for bilinear systems with autoregressive noises via the data filtering. *ISA Trans* 2024;147:337–49.
- [86] Xu L, Xu H, Wei C, Ding F, Zhu QM. The filtering-based recursive least squares identification and convergence analysis for nonlinear feedback control systems with coloured noises. *Internat J Sci Syst* 2024;55(16):3461–84.
- [87] Zhang X. Adaptive parameter estimation for a general dynamical system with unknown states. *Internat J Robust Nonlinear Control* 2020;30(4):1351–72.
- [88] Wei C, Zhang X, Xu L. Overall recursive least squares and overall stochastic gradient algorithms and their convergence for feedback nonlinear controlled autoregressive systems. *Internat J Robust Nonlinear Control* 2022;32(9):5534–54.
- [89] Xu L. Separable recursive gradient algorithm for dynamical systems based on the impulse response signals. *Int J Control Autom Syst* 2020;18(12):3167–77.
- [90] Zhang X, Yang EF. Highly computationally efficient state filter based on the delta operator. *Internat J Adapt Control Signal Process* 2019;33(6):875–89.
- [91] Xu L. Hierarchical multi-innovation generalised extended stochastic gradient methods for multivariable equation-error autoregressive moving average systems. *IET Control Theory Appl* 2020;14(10):1276–86.

- [92] Zhang X, Yang EF. State estimation for bilinear systems through minimizing the covariance matrix of the state estimation errors. *Internat J Adapt Control Signal Process* 2019;33(7):1157–73.
- [93] Wang YJ. Filtering-based iterative identification for multivariable systems. *IET Control Theory Appl*. 2016;10(8):894–902.
- [94] Wang DQ. Input-output data filtering based recursive least squares parameter estimation for CARARMA systems. *Digit Signal Process* 2010;20(4):991–9.
- [95] Zhang X. Hierarchical parameter and state estimation for bilinear systems. *Int J Syst Sci* 2020;51(2):275–90.
- [96] Cao Y, An Y, Su S, Xie G, Sun Y. A statistical study of railway safety in China and Japan 1990–2020. *Accid Anal Prev* 2022;175:106764.
- [97] Cao Y, Yang YR, Ma LC, Wen JK. Research on virtual coupled train control method based on GPC & VAPF. *Chin J Electron* 2022;31(5):897–905.
- [98] Sun YK, Cao Y, Li P. Contactless fault diagnosis for railway point machines based on multi-scale fractional wavelet packet energy entropy and synchronous optimization strategy. *IEEE Trans Veh Technol* 2022;71(6):5906–14.
- [99] Cao Y, Ji YS, Sun YK, Su S. The fault diagnosis of a switch machine based on deep random forest fusion. *IEEE Intell Trans Syst Mag* 2023;15(1):437–52.
- [100] Wang X, Su S, Cao Y, Wang XL. Robust control for dynamic train regulation in fully automatic operation system under uncertain wireless transmissions. *IEEE Trans Intell Transp Syst* 2022;(11):20721–34.
- [101] Cao Y, Ma LC, Xiao S. Standard analysis for transfer delay in CTCS-3. *Chin J Electron* 2017;26(5):1057–63.
- [102] Cao Y, Zhang ZX, Cheng FL, Su S. Trajectory optimization for high-speed trains via a mixed integer linear programming approach. *IEEE Trans Intell Transp Syst* 2022;23(10):17666–76.
- [103] Cao Y, Sun YK, Xie G, Li P. A sound-based fault diagnosis method for railway point machines based on two-stage feature selection strategy and ensemble classifier. *IEEE Trans Intell Transp Syst* 2022;23(8):12074–83.
- [104] Cao Y, Wen JK, Hobiny A, Li P, Wen T. Parameter-varying artificial potential field control of virtual coupling system with nonlinear dynamics. *Fractals* 2022;30(2):2240099.
- [105] Cao Y, Wen JK, Ma LC. Tracking and collision avoidance of virtual coupling train control system. *Alex Eng J* 2021;60(2):2115–25.
- [106] Sun YK, Cao Y, Ma LC. A fault diagnosis method for train plug doors via sound signals. *IEEE Intell Trans Syst Mag* 2021;13(3):107–17.
- [107] Sun YK, Cao Y, Xie G, Wen T. Sound based fault diagnosis for RPMs based on multi-scale fractional permutation entropy and two-scale algorithm. *IEEE Trans Veh Technol* 2021;70(11):11184–92.
- [108] Su S, Wang XK, Cao Y, Yin JT. An energy-efficient train operation approach by integrating the metro timetabling and eco-driving. *IEEE Trans Intell Transp Syst* 2020;21(10):4252–68.
- [109] Cao Y, Wang Z, Liu F, Li P, Xie G. Bio-inspired speed curve optimization and sliding mode tracking control for subway trains. *IEEE Trans Veh Technol* 2019;68(7):6331–42.
- [110] Cao Y, Sun YK, Xie G, Wen T. Fault diagnosis of train plug door based on a hybrid criterion for IMFs selection and fractional wavelet package energy entropy. *IEEE Trans Veh Technol* 2019;68(8):7544–51.
- [111] Su S, She JF, Li KC, Wang X, Zhou Y. A nonlinear safety equilibrium spacing based model predictive control for virtually coupled train set over gradient terrains. *IEEE Trans Transp Electrif* 2022;8(2):2810–24.
- [112] Su S, Zhu QY, Liu JQ, Tang T, Wei QL, Cao Y. Eco-driving of trains with a data-driven iterative learning approach. *IEEE Trans Ind Inf* 2023;19(3):7885–93.
- [113] Su S, Tang T, Xun J, Cao F, Wang YH. Design of running grades for energy-efficient train regulation: A case study for Beijing Yizhuang line. *IEEE Intell Trans Syst Mag* 2021;13(2):189–200.
- [114] Su S, Wang XK, Tang T, Wang G, Cao Y. Energy-efficient operation by cooperative control among trains: A multi-agent reinforcement learning approach. *Control Eng Pract* 2021;116:104901.
- [115] Chen MT, Hayat T. Iterative identification algorithms for bilinear-in-parameter systems with autoregressive moving average noise. *J Franklin Inst* 2017;354(17):7885–98.